HARSH G. BHUNDIYA

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ACADEMIC APPOINTMENTS:

University of Maryland, College Park **2026 – present**
Assistant Professor
Department of Aerospace Engineering

University of Southern California **2025 – 2026**
Postdoctoral Research Fellow
Department of Astronautical Engineering

EDUCATION:

Massachusetts Institute of Technology **2025**
Ph.D., Department of Aeronautics and Astronautics
Thesis: Rapid In-Space Assembly and Manufacturing of Large Reticulated Truss Structures
Advisor: Prof. Zachary C. Cordero
Committee: Prof. David Miller, Prof. Jack Wisdom, Prof. Shane Ross
Readers: Prof. Olivier de Weck, Prof. Sergio Pellegrino, Dr. Michael Marshall

California Institute of Technology **2020**
B.S., Mechanical Engineering, Minor: Aerospace Engineering

AWARDS AND HONORS:

- NASA Space Technology Graduate Research Fellow, **2023-2025**
- MIT School of Engineering Communication Fellow, **2022-2025**
- MathWorks Engineering Fellow, **2022-2023**
- Tau Beta Pi Fellow, **2022-2023**

JOURNAL PUBLICATIONS:

Superscript * denotes the corresponding author.

J1. **H.G. Bhundiya***, Z.C. Cordero, and M.A. Marshall, "Passive Gravity Gradient Capture during In-Space Assembly and Manufacturing," *AIAA Journal* (*accepted*).

J2. **H.G. Bhundiya**, M.A. Marshall, and Z.C. Cordero*, "Fabrication Time Diagrams for In-Space Manufacturing of Large Reticulated Structures," *Journal of Manufacturing Science and Engineering*, 146(12), 2024.

J3. J.Z. Zhang, **H.G. Bhundiya**, K.D. Overby, F. Royer, J.H. Lang, Z.C. Cordero*, W.F. Moulder, S.K. Jeon, and M.J. Silver, "Electrostatically Actuated X-Band Mesh Reflector with Bend-Formed Support Structure," *Journal of Spacecraft and Rockets*, 61(6), 2024.

J4. **H.G. Bhundiya**, Z.C. Cordero*, "Bend-Forming: A CNC Deformation Process for Fabricating 3D Wireframe Structures," *Additive Manufacturing Letters*, 6, 2023.

J5. **H.G. Bhundiya**, F. Royer, and Z. Cordero*, "Engineering Framework for Assessing Materials and Processes for In-Space Manufacturing," *Journal of Materials Engineering and Performance*, 31(2), 2022.

CONFERENCE PRECEEDINGS:

C1. **H.G. Bhundiya**, A. Soval, K. Albee, "More than a Manipulator: Planning Propellant-Free Attitude Maneuvers for Free-Floating Spacecraft," International Conference on Space Robotics (ISPARO) (Cologne, Germany), Nov 2026 (*accepted*).

C2. A. Soliman, S. Shaklan, J. Mather, E. Peretz, C. Gregg, M. Arya, M. Damiano, M. Kasper, J. Nousiainen, J. Taaki, J.L. Christiansen, A. Chandra, Z. Ahmed, **H.G. Bhundiya**, R. Dekany, J. Garcia-Mejia, J. Lopez, D. Miller, R. Morgan, S. Pellegrino, P. Plavchan, W. Sanchez, A. Sanghi, J. Sauder, A. Szentgyorgyi, R. Turner, A. Young, A. Zelakiewicz, H. Knutson, S. Jeon, D. Savransky, J. O'Meara, S. Brown, C. Lovis, M. Mayor, M. Silver, J. Arenberg, "Toward a Hybrid Ground-Space Exoplanet Observatory: Summary of Caltech Keck Institute for Space Studies (KISS) Report," SPIE Space Telescopes and Instrumentation (Copenhagen, Denmark), July 2026.

C3. **H.G. Bhundiya**, Z.C. Cordero, M.A. Marshall, "Passive Gravity Gradient Capture for In-Space Assembly and Manufacturing," AIAA/AAS Astrodynamics Specialist Conference (Boston, MA), Aug 2025.

C4. **H.G. Bhundiya**, Z.C. Cordero, "Radially Expanding Euler Paths for Assembly of Truss Structures," International Symposium on Space Technology and Science (Tokushima, Japan), Jul 2025.

C5. **H.G. Bhundiya**, Z.C. Cordero, M.A. Marshall, S. Mohan, D. Sternberg, and K. Lo, "Ground Testing of Spacecraft Attitude Dynamics During In-Space Assembly and Manufacturing," AIAA Scitech Forum (Orlando, FL), Jan 2025.

C6. **H.G. Bhundiya**, J.Z. Zhang, K.D. Overby, F. Royer, J.H. Lang, Z.C. Cordero, W.F. Moulder, S.K. Jeon, and M.J. Silver, "Electrostatically Actuated X-Band Mesh Reflector with Bend-Formed Support Structure," AIAA Scitech Forum (National Harbor, MD), Jan 2023.

C7. F. Royer, J.Z. Zhang, K.D. Overby, E.Y. Zhu, **H.G. Bhundiya**, J.H. Lang, and Z.C. Cordero, "Electrostatically Actuated Thin-Shell Space Structures," AIAA Scitech Forum (National Harbor, MD), Jan 2023.

C8. **(*Awarded 2022 AIAA Spacecraft Structures Best Paper*) H.G. Bhundiya**, F. Royer, and Z. Cordero, "Compressive Behavior of Isogrid Columns Fabricated with Bend-Forming," AIAA SciTech Forum (San Diego, CA), Jan 2022.

C9. **H.G. Bhundiya**, M. Hunt, and B. Drolen, "Measurement of the Effective Radial Thermal Conductivities of 18650 and 26650 Lithium-Ion Battery Cells," NASA Thermal and Fluid Analysis Workshop (Galveston, TX), Aug 2018.

INTELLECTUAL PROPERTY:

P1. **H.G. Bhundiya**, Z.C. Cordero, "Computer Numerical Control (CNC) Deformation Process for Forming 3D Wireframe Structures." US Patent Application No. 18/147,674, Dec 2022.

INVITED TALKS:

T1. ASTM International Conference on Advanced Manufacturing, "Design for In-Space Manufacturing: A Framework for Efficient and Rapid Construction of Space Structures," Sep 2026.

T2. ASME Aerospace Structures, Structural Dynamics, and Materials Conference, "Towards Rapid and Efficient Assembly and Manufacturing of Large Structures in Space," Jun 2026.

T3. University of Illinois Urbana-Champaign Workshop on In-Space Manufacturing of Resilient Structures, "High-Rate Manufacturing of Large Precision Space Structures," May 2026.

T4. Johns Hopkins University Applied Physics Laboratory Lunar Surface Innovation Consortium, "High-Level Planning for Multi-Agent Construction of Large Space Structures," Virtual, Dec 2025.

T5. The Aerospace Corporation, "Rapid In-Space Assembly and Manufacturing of Next-Generation Space Structures," Los Angeles, CA, Dec 2025.

T6. California Institute of Technology, ME10 Class Guest Lecture, "Constructing Large Structures in Space," Pasadena, CA, Nov 2025

T7. NASA Langley Research Center, "Rapid In-Space Assembly and Manufacturing of Next-Generation Space Structures," Hampton, VA, Aug 2025.
T8. Johns Hopkins University Applied Physics Laboratory, "Rapid In-Space Assembly and Manufacturing of Next-Generation Space Structures," Laurel, MD, Jul 2025.
T9. NASA Goddard Spaceflight Center, "Rapid In-Space Assembly and Manufacturing of Next-Generation Space Structures," Greenbelt, MD, Jul 2025.
T10. AIAA Emerging Spacecraft Structures Technology Workshop, "Passive Gravity Gradient Capture for In-Space Assembly and Manufacturing," Boulder, CO, Jun 2025.
T11. MIT Small Satellite Collaborative Seminar, "Fabrication Time Diagrams for In-Space Manufacturing of Reticulated Structures," Boston, MA. Oct 2024.
T12. AIAA Emerging Spacecraft Structures Technology Workshop, "Fabrication Time Diagrams for In-Space Manufacturing of Reticulated Structures," Boston, MA. Jul 2024.
T13. Indian Institute of Technology Gandhinagar, "In-Space Manufacturing of Large Electrostatically-Actuated Mesh Reflectors," Gujarat, India, Jan 2024.
T14. AIAA Emerging Spacecraft Structures Technology Workshop, "Spacecraft Dynamics during In-Space Manufacturing," Stanford, CA, Aug 2023.
T15. U.S. National Congress of Computational Mechanics, "Bend-Forming: A CNC Deformation Process for Fabricating 3D Wireframe Structures," Albuquerque, NM, Jul 2023.
T16. AFRL Space Vehicles Directorate, "In-Space Manufacturing of Large Electrostatically-Actuated Mesh Reflectors," Albuquerque, NM, Jul 2023.

MENTORING AND ADVISING:

Ph.D. Researchers:

- José Morel UMD Fall 2026 – present
- Sam Ondrusek UMD Fall 2026 – present

Ph.D. Committee Member:

- Grazchella Vicente UMD Fall 2026 – present

B.S. Researchers:

- Spencer Bradley UMD Fall 2026 – present

PROFESSIONAL SERVICE:

- **Technical Manuscript Reviewer:** Acta Astronautica, Journal of Spacecraft and Rockets, Journal of Aerospace Information Systems, AIAA Scitech Forum
- **Participant:** Keck Institute of Space Studies Workshop, "Hybrid Space-Ground Observatories: Revolutionizing the Search for Earth-like Exoplanets," Mar 9-13, 2026
- **Co-Organizer:** IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) Workshop, "Space Exploration and Sustained Operations Beyond Earth," Sep 27, 2026
- **Member:** American Institute of Aeronautics and Astronautics (AIAA) Spacecraft Structures Technical Committee
  - Deputy Technical Discipline Chair (2026-2027)
  - Technical Discipline Chair (2027-2028)

TEACHING EXPERIENCE:

University of Southern California

Spring 2026 Future Faculty Teaching Institute (*Participant*)

OUTREACH ACTIVITIES:

- **Mentor and Judge**, Consortium for Space Mobility and ISAM Capabilities. Mentored student teams for C3 Capstone Challenge (2025-26).
- **Educational Blogger,** maintains an educational blog titled "Harsh World of Mechanics" focused on everyday applications of mechanics of materials and structures (2021-present)
- **Lecturer,** Cambridge Science Festival (2023). Gave a lecture titled "Manufacturing Large Structures in Space" at the Cambridge Science Festival hosted by the MIT Museum.
- **Instructor,** MIT Educational Studies Program (2022-2024). Taught a biannual class on "Geometry and Beauty of Soap Bubbles" to >100 middle school and high school students.